AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2011

General Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are THIRTEEN questions. Answer TEN questions. Choices are specified inside the paper

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use colour pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

Blue and yellow paints are mixed in the ratio 3:4 to make a colour of Ambreen's choice. If 18 litres paint is required to paint the exterior of her apartment, then estimate the total cost of paint by using the given table.

Paint Colour	Price (per litre)
Blue	Rs. 800
Yellow	Rs. 820

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 4 Marks)

a. Mrs. Farooq paid Rs.6,825 and Rs.12,775 as Zakat on her annual savings and gold jewellery, respectively. Find her annual savings and value of the jewellery in Rupees. Also find the total amount for which Zakat is due.

b. Mr. Nawaz divided his property among his four daughters and four sons. Each daughter gets Rs.2,25,000 which is $8\frac{1}{3}\%$ of the total property. If each daughter gets half the share of a son, then find the value of the property and the total share of all four sons in rupees.

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 5 Marks)

a. A stationer bought 20 dozens of geometry boxes at Rs. 36 each. He sells 75 of them at Rs.45 each. What will be the selling price of each of the remaining boxes in order to make a profit of 20% on the overall sale?

b. Mr. Tahir bought a second hand car and a motor cycle for Rs. 3,50,000 and Rs. 40,000, respectively. Later, he sold the car at a loss of 15% and the motor cycle at a profit of 9%. Find the net profit or loss percentage in these two sales.

Q.4. (Total 3 Marks)

A simple interest of Rs.3000 is offered to Mr. Najam on the investment of certain amount of money for 8 months at 12% per annum. Find the principal amount of the investment of Mr. Najam.

Q.5. (Total 5 Marks)

A furniture shop sells mattresses of two companies A and B at the rate of Rs.9,800 and Rs.9,000, respectively. The commission for salespersons at the shop is 5% on each sale. The sales of two salespersons in a certain week are given in the following table:

	Number of mattresses sold of company A	Number of mattresses sold of company B
Zahid	6	4
Wajid	4	8

Find the difference in commission of Mr. Zahid and Mr. Wajid in rupees.

(ATTEMPT EITHER Q.6. OR Q.7.)

Q.6. (Total 5 Marks)

a. Simplify $\frac{\sqrt[3]{16x^2y^7}}{\sqrt[3]{2x^5y}}$. (3 Marks)

b. Simplify $105\times\left(2.12\times10^{-4}\right)$ and write your answer in scientific notation. (2 Marks)

(ATTEMPT EITHER Q.6. OR Q.7.)

Q.7. (Total 5 Marks)

Apply properties of logarithm to find the value of x for $\log_4(x-1)+2\log_4 3=\frac{1}{2}$.

Q.8. (Total 4 Marks)

Insert two arithmetic means between 2 and 7.

(ATTEMPT EITHER Q.9. OR Q.10.)

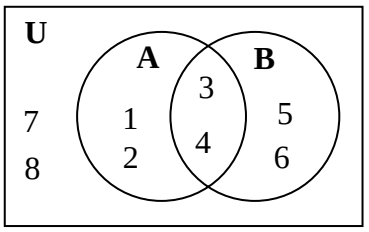
Q.9. (Total 5 Marks)

If $A=\{1,3,5,7,9\}$ and $B=\{7,9,10,11\}$, then find $(A\cup B)-(A\cap B)$ and $(A\cap B)-(A\cup B)$. Does the result of $(A\cap B)-(A\cup B)$ remain the same for any two sets *A* and *B*?

(ATTEMPT EITHER Q.9. OR Q.10.)

Q.10. (Total 5 Marks)

For the given Venn diagram



i. shade $(A\cup B)'$.

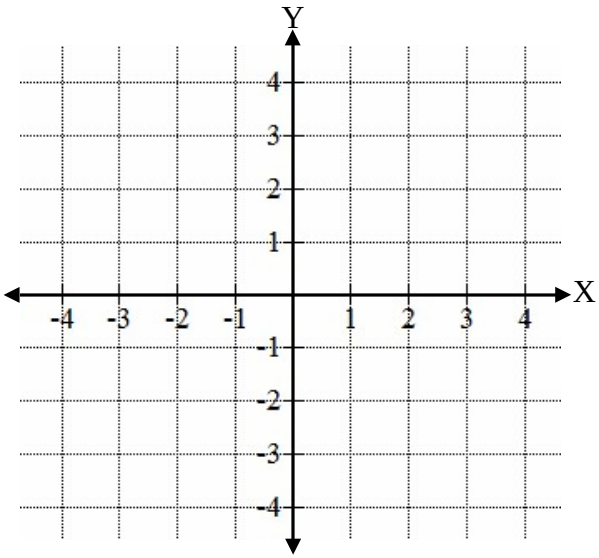
ii. write the elements of universal set U.

iii. find $A'\cap B'$.

iv. compare the results of i and iii and write a relation between $(A\cup B)'$ and $A'\cap B'$.

Q.11. (Total 5 Marks)

a. Locate A (0, 3), B (0, –1), C (2, –1) and D (2, 3) on the following graph paper. Join these points to form a quadrilateral ABCD and identify the quadrilateral. (3 Marks)



b. Complete the following table for the linear equation $5x - 3y = 4$. (2 Marks)

x		5
y	5	

(ATTEMPT EITHER Q.12. OR Q.13.)

Q.12. (Total 6 Marks)

The Quality Assurance Department of a factory measured the diameter of 8 randomly selected steel balls. The measurements are shown below:

Sample	1	2	3	4	5	6	7	8
Diameter (in mm)	17.5	17.6	18.3	18.5	18.5	18.6	19.1	19.3

Calculate

i. the average diameter of the balls.

ii. the variance in diameter of the balls.

iii. the standard deviation in the diameter of the balls.

(ATTEMPT EITHER Q.12. OR Q.13.)

Q.13. (Total 6 Marks)

Following are the gas bills of Mr. Sohail's house for year 2010:

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
Bills (in Rs.)	171	160	197	140	118	171	157	220	173	192	293	317

i. Find the range of the data.

ii. Calculate average bills for January to June and July to December and decide which six months are less expensive in terms of gas consumption.

iii. Find the median of the data.

END OF PAPER

Please use this page for rough work

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